

Lab 02

Data Structures

BS DS/CY Fall 2025 Morning/Afternoon

Task 1

Define a function which prints the Row-Major based ND-array formula against a given number of dimensions.

For Example;

If the function is called as “printND(3)” then the following should be displayed on console:

$$I_1U_2U_3 + I_2U_3 + I_3$$

for “printND(1)”

$$I_1$$

for “printND(2)”

$$I_1U_2 + I_2$$

Where $I_1, I_2, I_3, \dots, I_N$ represents the index set and $U_1, U_2, U_3, \dots, U_N$ represents the dimension set.

Task 2

Requirements:

Even though the multidimensional array is provided as a standard data object in C++, it is often useful to define your own class for multidimensional arrays. This gives a more robust class that:

- Does require the index set in each dimension to start at any number (€ Set of Integers).
- Selects range of each dimension of the array during runtime.
- Provide mechanism to determine the size of array.

No specific data structure is being told to you for this task so intelligently do a proper structure selection yourself and when feels confident then also discuss your structure with Instructor before start coding.

Sample Program Run

```
int dim-size[3]={5,3,10};

NDArray arr(3, dim-size);

int index-set[3]={4,2,8};
cout<<arr.calculateIndex( indexset )<<endl;

arr.setValue( index-set , val);

cout<<arr.getValue(index-set);
```

Task 3: Deals with buffered writing!

Objective:

- The main objective of this is to experience the effect on performance (time) with the use of buffered reading/writing.
- Getting the basic concept of indexing and its advantage on performance (time).
- Hands on STL.

Prerequisite Setup:

1. Define a struct 'Student' as follows:

```
struct Student
{
    int roll;
    char name[30];
    Student():roll(0)
    {
        strcpy(name,"none");
    }
};
```

2. The following code stores 10 million records of 'Student' in a file in binary mode pass N = 10000000

```
void addToStudentUnBuffered(int N)
{
    ofstream ofs("studentdatabase.txt",ios::binary|ios::out);
    Student s;
    for (int i=1; i<N; i++)
    {
        s.roll = i;
        ofs.write((char*)&s,sizeof(Student));
    }
    ofs.close();
}
```

3. Write the following code in main: [Required header files **stdio.h and **windows.h**]**

```
SYSTEMTIME systime;
cout<<"\nWriting Records to File one by one";
GetLocalTime(&systime);
cout<<endl<<systime.wHour<<":"<<systime.wMinute<<":"<<systime.wSecond<<":"<<systime.wMilliseconds;
addToStudentUnBuffered();
GetLocalTime(&systime);
cout<<endl<<systime.wHour<<":"<<systime.wMinute<<":"<<systime.wSecond<<":"<<systime.wMilliseconds;
```

Can you improve the performance of addToStudentUnBuffered ?

Idea is to go for buffered writing. Name your function as addToStudentBuffered

Task 4: Deals with buffered reading!

Write a function which display all the Student records stored in "studentdatabase.txt" in Task-3.

[Maybe you noticed or not: its 343 MB file]

Note: Write two versions of display all records as follows:

- ➔ readAllRecordsUnBuffered() //:It reads records from file one by one i.e. unbuffered.
- ➔ readAllRecordsBuffered() //:It reads records from file in a buffer.

Then observe the response time of both functions by getting the system time.